Procedural Generation of Solvable Levels in MARL Environment

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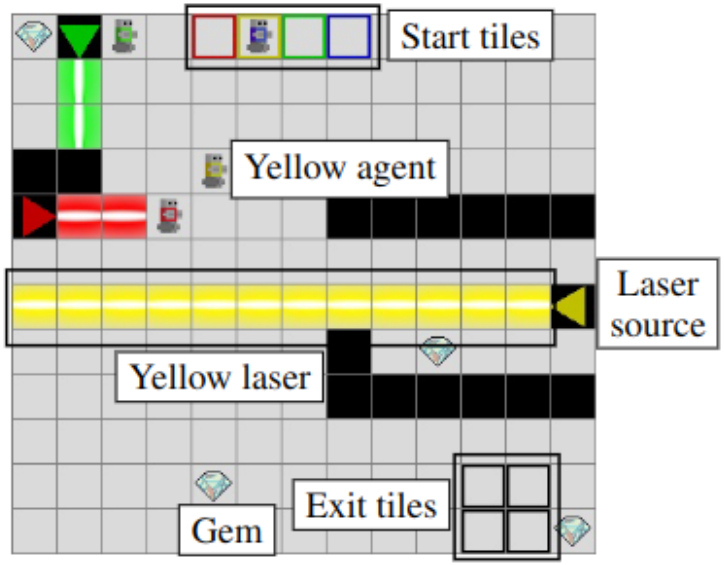
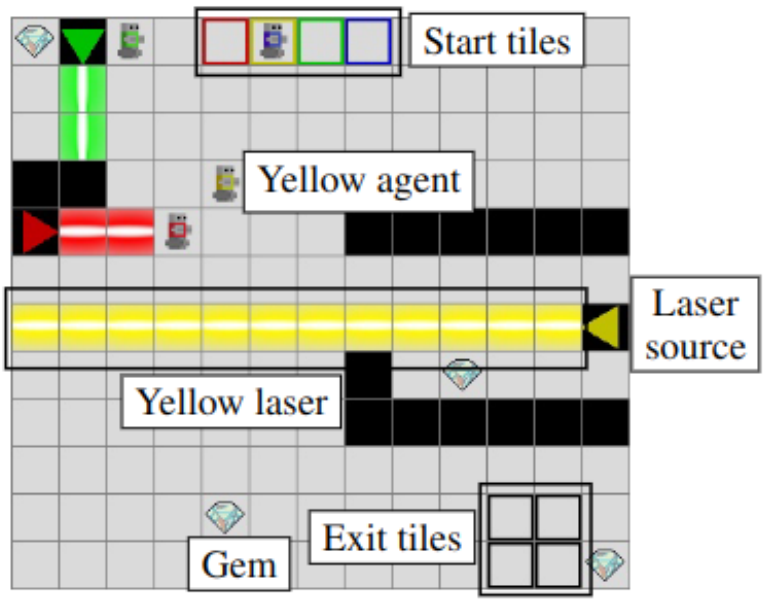


Context & Motivation

Laser Learning Environment (LLE)

- 2D grid-based fully cooperative multi-agent puzzle
- Each agent must reach its exit tile **simultaneously** with all others
- Laser beams are passable only by an agent of the **matching color**
- Reward signal is sparse: zero credit for intermediate coordination steps

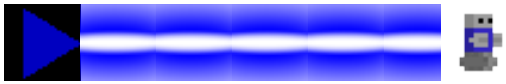


Research problem

- MARL training requires **diverse, solvable** levels with genuine cooperative structure
- Hand-crafted levels are expensive, non-scalable, and inherently biased

Objective

Procedurally generate LLE levels that are **solvable**, **learnable** by RL agents, and **structurally require** inter-agent cooperation.



LLE Solvability

Single-agent case

- Reduces to graph reachability: solvable in polynomial time via BFS, DFS, ...
- Laser avoidance is a static constraint; no interaction between agents

Multi-agent case — structurally harder

- Agents **actively modify the environment**: blocking a laser opens a path for another agent
- Agent actions are **mutually dependent**: feasibility of a move depends on others' positions
- Hard synchronization constraint: all agents must exit **simultaneously**
- Naive joint-state search is exponential in the number of agents

Theorem (NP-membership)

A valid joint execution trace is a polynomial-size certificate, checkable in polynomial time — hence LLE Solvability $\in$ NP.

Open question: Is LLE Solvability NP-hard?

Background: Satisfiability Problem

Definition (SAT)

Given a Boolean formula ϕ over variables $x_1, \ldots, x_n$, decide whether there exists an assignment $\alpha : \{x_1, \ldots, x_n\} \to \{0, 1\}$ such that $\phi(\alpha) = \top$.

Definition (CNF)

A formula is in conjunctive normal form if it is a conjunction of one or more clauses, where a clause is a disjunction of literals

$$F = \bigwedge_{i=1}^{m} \left(\bigvee_{j=1}^{k_i} l_{i,j} \right)$$

Theorem (Cook–Levin)

SAT is NP-complete.

Solver

Definition (Reduction)

A **reduction** from a problem A to a problem B is a transformation that maps any instance of A to an instance of B such that the answer is preserved. In particular, solving B allows us to solve A.

Variables

- $a_{c,x,y,t}$: agent with color c at position (x, y) after t time steps
- $b_{c,d,x,y,t}$: beam of color c with direction d at position (x, y) after t time steps
- $l_{c,x,y,t}$: laser of color c active at position (x, y) after t time steps

Constraints

Initialization

- starting tiles
- laser sources

Movements

- legal movements
- unique position
- no overlap between agents
- exit tiles

Lasers

- beam propagation
- no step on active laser
- link between beams and lasers

Agents movements

Legal movement:

$$\bigwedge_{c \in C} \bigwedge_{t \in T} \bigwedge_{(x,y) \in P} a_{c,x,y,t} \rightarrow \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t+1}$$

Agents movements

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At most one agent per tile:

$$\bigwedge_{c \in C} \bigwedge_{t \in T} \bigwedge_{\substack{(x_1,y_1),\\(x_2,y_2) \in P\\(x_1,y_1) \neq (x_2,y_2)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

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Global

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Local

Agent should come from a legal movement:

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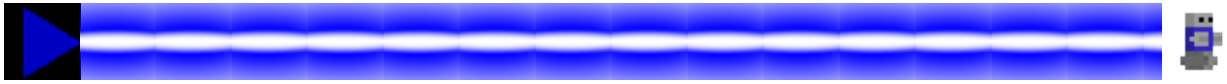
At most one future position:

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{\substack{(x',y'),(x'',y'')\in \text{next}(x,y)\\(x',y')\neq(x'',y'')}} \neg a_{c,x',y',t} \vee \neg a_{c,x'',y'',t}$$



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At most one agent per tile:

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$$f(n) = \binom{n}{2} = \frac{(n-1)n}{2} \in O(n^2)$$

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$$g(n) = (f(5)+1)n = 11n \in O(n)$$

Agents movements

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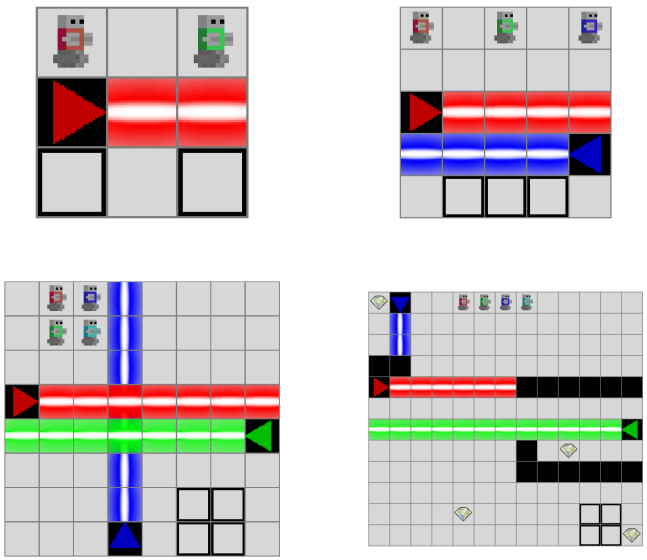
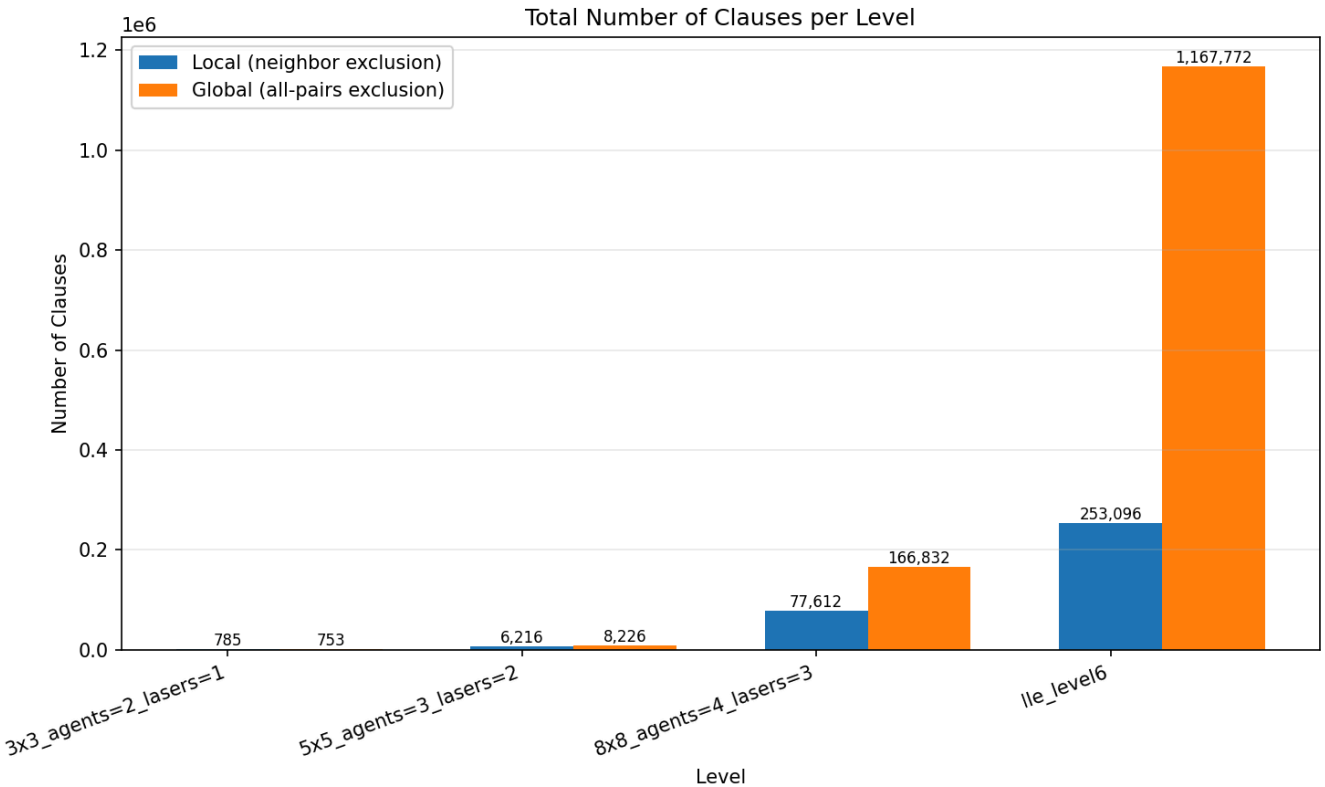
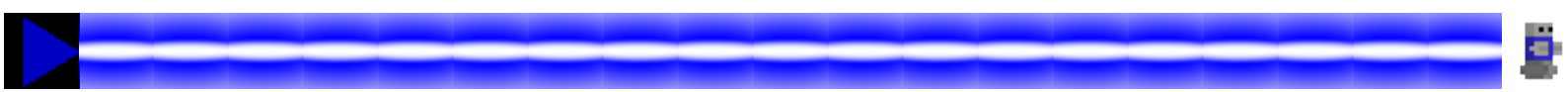
$$g(n) = (f(5)+1)n = 11n \in O(n)$$

$f(n) < g(n)$ when $n < 23$ $\quad$ $f(n) = g(n)$ when $n = 23$ $\quad$ $f(n) > g(n)$ when $n > 23$

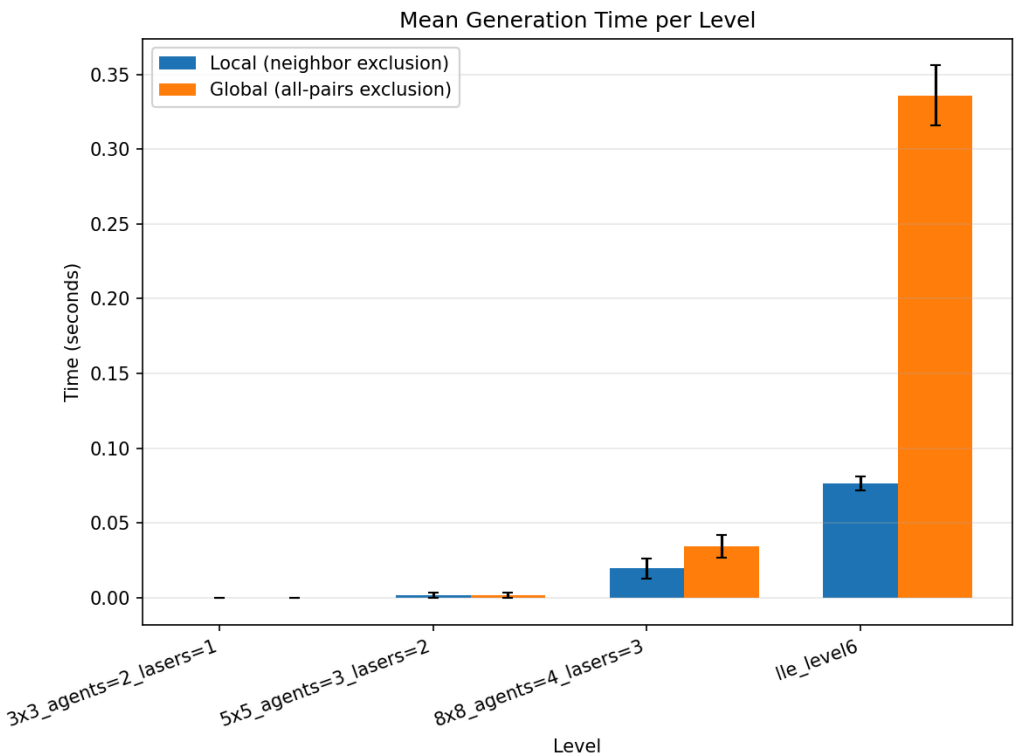
Plots clauses number difference

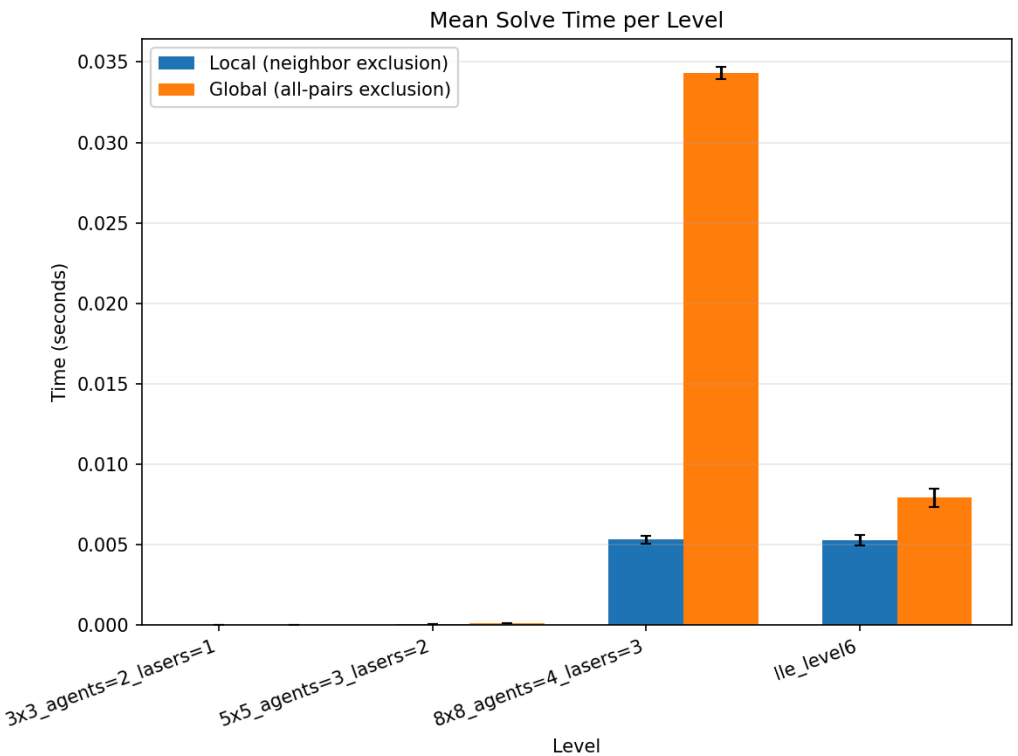






Plots time difference





Future Work

Solver with cooperation constraints

- Extend the SAT encoding to include cooperation-specific constraints
- Use the solver to analyze the cooperative structure of levels and identify key coordination points

Procedural Level Generation

- Explore the use of SAT solvers for generating levels
- Develop algorithms to generate solvable levels with specific properties (e.g., difficulty, required cooperation patterns)

Experimental Validation

- Train state-of-the-art MARL agents on generated curricula
- Compare convergence and generalization against hand-designed baselines

Complexity

- Formal reduction from a known NP-hard problem to LLE Solvability and Establish NP-completeness of the decision problem
- Or find a polynomial-time algorithm for LLE Solvability, which would be a surprising result

Summary

Topics covered

1. **Context & Motivation** — LLE, cooperative MARL, PCG
2. **LLE Solvability** — hardness analysis, NP-membership
3. **SAT Background** — CNF, Cook–Levin theorem
4. **SAT Encoding** — polynomial reduction from LLE to SAT
5. **Movement Constraints** — comparaison of two constraint formulations
6. **Empirical Results** — solver performance across levels
7. **Future Work** — solver with cooperation constraints, procedural generation, experimental validation, complexity analysis